Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 3.5+ years of applied LLM engineering and 8+ years of production systems, specializing in **agentic AI architecture, RAG platform design, and eval-first LLM development** for real estate and compliance domains. Built AgentProbe — an open-source multi-agent AI evaluation framework (LangGraph, 5-metric LLM-as-judge eval engine, CI/CD quality gates, regression drift detection). Built Leasel — an agentic LangGraph RAG pipeline for real estate document analysis (93.6% precision, 87.9% F1). Drove technical direction for Entrata's first production AI pipeline (PyTorch, OpenAI API) across 1.6M real financial records — shipped to production and later productized by the team. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 -- Aug 2024

Trainee → Associate Engineer → Software Engineer

- Drove technical direction for Entrata's **first production AI pipeline** (Python, PyTorch, OpenAI API) across 1.6M real financial records — designed the architecture, built output validation and quality gates, delivered to production, mentored the team through adoption; later productized into a customer-facing feature.
- Led cross-functional delivery of **20+ production modules** (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users across 1M+ property records — set technical direction, reviewed 300+ PRs, mentored 5+ engineers; 99.5% uptime over 6+ months.
- Built **affordable housing compliance automation** (LIHTC/Tax Credit, Section 8/HCV, HUD) — income eligibility, AMI calculations, recertification workflows across 100K+ unit portfolios; eliminated 80% of manual workload; zero compliance violations.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); improved query performance 30% across 200+ SQL queries; cut execution time 50% via Docker containerization.

George Mason University

Aug 2024 -- Aug 2025

Research Associate — Data Engineering · Statistical Modeling · Anomaly Detection

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries; applied Isolation Forest to automate anomaly detection across 3 research pipelines — surfaced anomalies manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership — improved research comprehension across two cohorts by 40%; translated complex model outputs into decisions for non-technical stakeholders.

PROJECTS

AgentProbe: Agentic AI Evaluation Framework

Open Source

Python · LangGraph · LangChain · FastAPI · ChromaDB · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Designed and built a **reusable LLM evaluation framework** — 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance), CI/CD quality gates, regression drift detection, and latency tracking across concurrent agent test suites; React dashboard for per-run observability.
- Architected a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (OpenAI, Anthropic, Groq, Ollama) via 15-endpoint FastAPI backend — modular, reusable agentic patterns deployable across products.

Leasel: Agentic RAG Platform for Real Estate Document Analysis

Python · LangGraph · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a production-grade agentic LangGraph RAG pipeline auditing 1,240 residential lease PDFs for compliance gaps — **93.6% precision, 82.9% recall, 87.9% F1** across 10 rules, 5 US states + federal standards; deterministic rules engine keeps every decision fully auditable.
- Designed an offline evaluation harness (pass@k: pass@1 81.7%, pass@3 88.3%) with regression gate and LLM-as-judge scoring for both accuracy and reasoning quality — same eval discipline the JD defines as a core leadership responsibility.

AI on FHIR: NLP Platform for Regulated Medical Records

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker — github.com/venkatalakshmiparimala/AI_on_FHIR

- Built an NLP platform translating natural language queries to FHIR APIs using LLMs, BERT, and semantic embeddings — **HIPAA-compliant RBAC**, ~60% reduction in retrieval time; demonstrates regulatory AI in a heavily compliance-constrained domain.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, HTML/CSS

AI Architecture & Agents: LangGraph, LangChain, Multi-Agent Orchestration, RAG Pipelines, LLM-as-Judge, LLM Evaluation Frameworks, AI Observability, Prompt Engineering & Versioning, ChromaDB, Vector Databases, Semantic Search, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, REST APIs

ML & Data: PyTorch, Scikit-learn, BERT, spaCy, Embeddings, NLP, Anomaly Detection, Feature Engineering, Statistical Modeling, NumPy, Pandas, Tableau, Power BI

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), Playwright, Microservices, ETL, Data Pipelines, RabbitMQ, PostgreSQL, MySQL, Git, Linux, Agile/Scrum